

What if a hummingbird wears a tiny backpack?

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Abstract

Scientists often use tiny tools to study wild animals. These tools can show where animals go and how they move. But if the tools aren't safe for the animals, then we aren't doing our job properly!

We studied Black-throated Mango hummingbirds in Colombia. These birds are small, fast, and full of energy. We wanted to know if wearing a tiny radio backpack changed how they spent their time.

Each bird had a tiny ID chip, like a name tag. We tested

each bird in two ways. The first time, the bird wore the radio backpack. The second time, it did not. The clearest change in the birds' behavior was preening. Birds with backpacks used their beaks to touch or pull at the straps more often. Birds with backpacks also flew a little less and ate a little more, but those results were not as consistent. No birds got caught in plants while wearing the backpack.

Our study shows that scientists should test tracking tools with great care before using them in the wild.

Introduction

Hummingbirds are amazing creatures. They can hover in the air, fly backward, and drink nectar from flowers while their wings beat very fast. But hummingbirds are also very small. A Black-throated Mango hummingbird weighs only about 8 grams. That is less than two teaspoons of sugar! A hummingbird can eat more than three times its own body weight in nectar each day just to keep moving. And they move fast! We know they can reach up to 31 miles per hour in forward flight. So how can scientists study such small, speedy birds?

Scientists often use **biologging** to study animals. Biologging means attaching tiny electronic tags to an animal to collect information about what it gets up to. It can show where an animal goes or how it moves. **These tools help scientists protect animals and their homes. But the tools must be safe.** They should not stop an animal from flying, eating, or resting.



Biologging helps scientists protect animals and their homes. But we must test tracking tools with great care before using them in the wild!

Photo: Samantha-Lynn Martinez

This is a big challenge! For a tiny bird, even a small **radio backpack** can feel like a heavy hiking backpack. Imagine if you had to go through your whole day running, jumping, and eating while wearing a bulky pack. It might change

how much you move. It would probably also change how often you need to rest.

We decided to test a small radio backpack. We asked: does it change how hummingbirds spend their time?

Methods

We worked with Black-throated Mango hummingbirds in Colombia, South America. To make sure we knew who was who, each bird received a tiny ID chip. Just like a name tag for a bird!

Then we tested a small radio transmitter. The birds wore it like a backpack. We call it the radio backpack. The birds flew in a large flight tent, so we gave each bird time to get used to the tent (Fig. 1). We tested each bird two times, filming it both times.

① The first time, the bird wore the radio backpack.

② The second time, it did not. The bird flew naturally with nothing on its back. This helped us compare each bird to itself.

We made a **time budget** for each bird. We measured flying, feeding, **perching**, and **preening**.

We also tested whether the backpack could get caught on plants. We call this **entanglement**. We watched birds in a tent with plants to check for this risk.

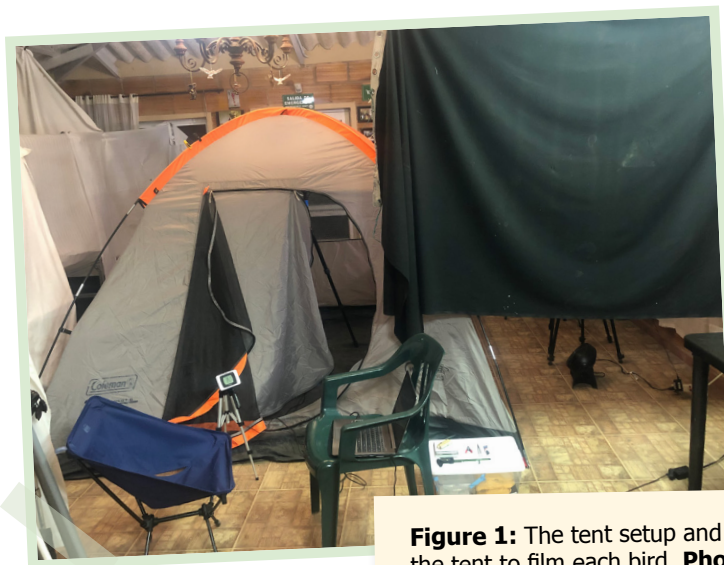


Figure 1: The tent setup and camera used inside the tent to film each bird. **Photo:** Alyssa J. Sargent

Results

The radio backpack did seem to affect the hummingbirds, especially in one specific way.

The "Itchy Sweater" Result

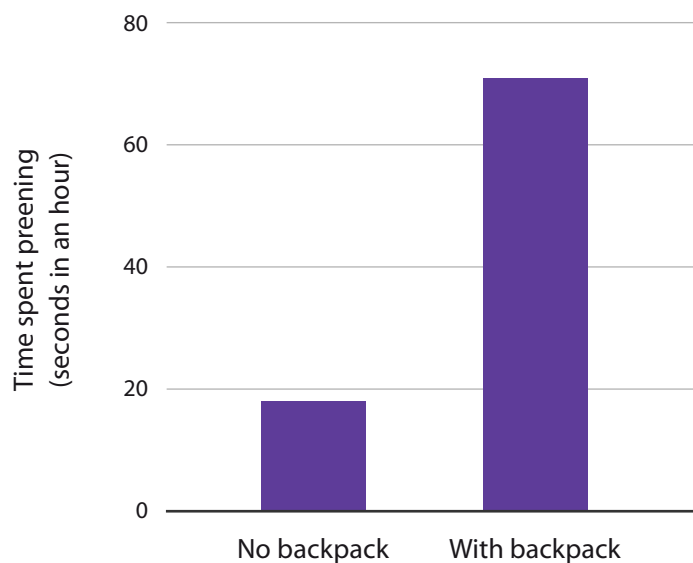
The clearest change was preening. **Birds wearing backpacks spent more time preening than birds without backpacks** (Fig. 2). This suggests that the birds noticed the harness. They used their beaks to touch or pull at the straps. It was a little like when you fidget with a new sweater that feels itchy.

Flying and Feeding Time

On average, birds with backpacks flew a little less than they did without them. They also spent more time feeding. But these results were not as clear as the preening result. The change was also not the same for every bird.

Safety First

The best news was that no birds experienced entanglement. Even in a tent with plants, the backpacks did not get caught on leaves or branches during our test.



What happened to preening time when the hummingbirds wore the radio backpack? What does this suggest?

Figure 2: Average time the hummingbirds spent preening with and without the radio backpack.

Discussion

Our study shows why careful testing matters. A tiny radio backpack can help scientists learn about hummingbirds. **Discovering more about their behavior can help us conserve these amazing birds. But the backpack may also change what the birds do.** If a bird preens more, it may be feeling the device on its body. This does not mean radio backpacks are always bad. **It means scientists should test them before using them in the wild.** If a bird spends extra time preening its harness, it may be using up precious energy.

Our study also had limits. The birds were in a flight tent. A tent is useful for science because it is easier to watch the birds. But it is not the same as a forest or garden. Wild birds may behave in other ways. Still, our test gave us useful information. The birds could fly, feed, perch, and move around. They did not get stuck in plants. Good scientific research should help us learn. But it should also protect the animals we study.

Conclusion

Next time you see a hummingbird zipping through your garden, remember how hard they work! If you want to help them, you can hang a feeder. Make sure to keep it clean so the birds stay healthy. When you watch wildlife, always be gentle and keep your distance. Watch birds without

touching them. Notice what they eat, where they rest, and how they move. Small animals have big needs, so even small actions can help. When we learn about wildlife with care, we become better neighbors to the animals around us.

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Glossary of Key Terms

Biologging - attaching a tiny electronic tag to an animal to collect information, such as where it goes or how it moves.

Entanglement - getting caught or tangled in something, such as plants, string, or netting.

Perching - sitting or resting on a branch, feeder, or other surface.

Preening - when a bird uses its beak to clean, fix, or arrange its feathers.

Radio backpack - a small radio transmitter that a bird wears on its back with a harness. Scientists can use it to help track the bird.

Time budget - a record of how much time an animal spends doing different activities, such as flying, feeding, perching, or preening.

Check your understanding

- 1 Why did each bird get a tiny ID chip?
- 2 Why did we test each bird with and without the radio backpack?
- 3 Do you think scientists should use radio backpacks to study hummingbirds? Why or why not?
- 4 Why were we concerned about entanglement?
- 5 Create a simple time budget for an animal you can observe, such as a bird, dog, cat, or insect. What behaviors could you measure?

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